Uniwersytet Wrocławski

Recommender Systems

Intro, Basics, Matrix Factorization

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Recommender Systems

Recommender Systemts

We are given:

- a set of N items $\mathcal{I} = \{i_1, i_2, \ldots, i_N\}$,
- a set of M users $\mathcal{U} = \{u_1, u_2, \ldots, u_M\}$,
- historical data $\mathcal{D}$ consisting of user-item interactions:
  - $\mathcal{D} = \{(u, i, r, t) :$
    user u interacted with item i at time stamp t with rating $r\}$,
  - $\mathcal{D} = \{(u, i, r) :$ user u interacted with item i with rating $r\}$,
  - $\mathcal{D} = \{(u, i, t) :$ user u interacted with item i at time stamp $t\}$,
  - $\mathcal{D} = \{(u, i) :$ user u interacted with item $i\}$,
- In matrix approaches we will use the notation r_{ui} for the rating of user u for item i and $\mathbf{R}$ for the matrix of all ratings.

The task of a Recommender System (RS) is to provide a list of K recommended items that meet the user's interest based on historical behavior of this and other users.

Data

- Interactions
  - Implicit vs Explicit feedback
  - Binary, Integer, Float
- Metadata
  - user and item metadata
  - multi-modal

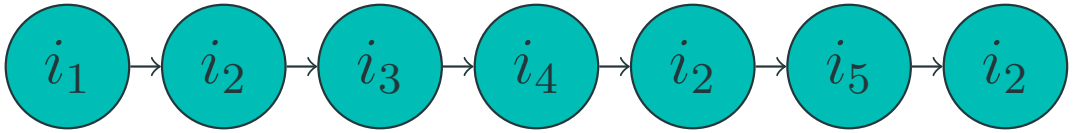
Example dataset:
https://cseweb.ucsd.edu/ jmcauley/datasets.html#amazon_reviews

Data representations

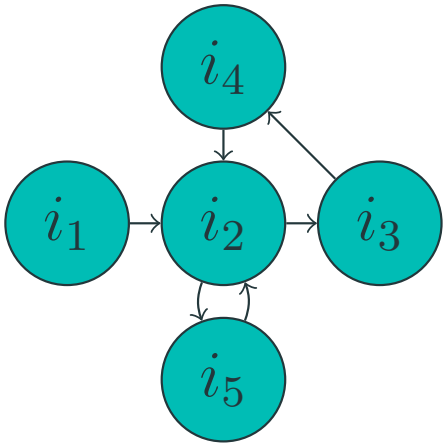
- matrix

$$\begin{bmatrix} 0 & \mathbf{1} & 0 & \mathbf{1} & \mathbf{1} & 0 & \mathbf{1} & \mathbf{1} & 0 & 0 \\ 0 & 0 & \mathbf{1} & 0 & \mathbf{1} & \mathbf{1} & 0 & 0 & \mathbf{1} & \mathbf{1} \\ 0 & 0 & 0 & \mathbf{1} & 0 & 0 & \mathbf{1} & 0 & 0 & 0 \\ 0 & \mathbf{1} & 0 & 0 & 0 & 0 & 0 & \mathbf{1} & 0 & 0 \\ 0 & \mathbf{1} & 0 & 0 & 0 & 0 & \mathbf{1} & 0 & 0 & 0 \end{bmatrix}$$

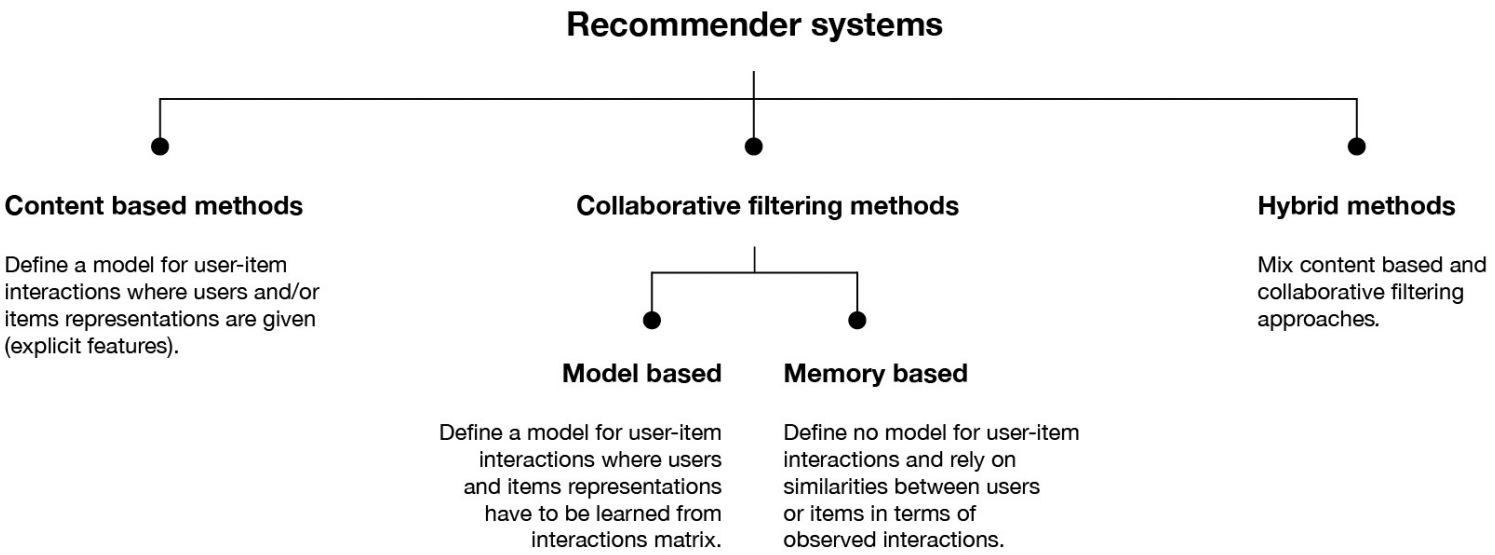
- sequence



- graph



Taxonomy



Source: https://medium.com/data-science/introduction-to-recommender-systems-6c66cf15ada

Approaches to Recommender Systems

Matrix Factorization

In Matrix Factorization approach:

- $\mathbf{P} \in R^{M \times D}$ - user embeddings ($\mathbf{p_u}$ - vector corresponding to user u);
- $\mathbf{Q} \in R^{N \times D}$ - item embeddings ($\mathbf{q_i}$ - vector corresponding to item i)

We optimize them so as to obtain:

- $r_{ui} \approx \mathbf{p_u} \times \mathbf{q_i}$,
- $\mathbf{R} \approx \mathbf{P} \times \mathbf{Q}^\top$.

Look for more:
https://surprise.readthedocs.io/en/stable/matrix_factorization.html#surprise.pre

Transform the matrix to a personalized ranking

To serve recommendations:

- calculate the scores for all items,
- omit items from training data,
- order ratings based on their predicted ratings / scores,
- we recommend K items with highest predicted ratings / scores.

Follow:
https://colab.research.google.com/drive/1uD5Zo9W1Pb4gtmZ9rKDR2osm56ww

Evaluation fo RS

Common metrics

- Recall
- Precision
- AUC
- Hit-rate
- MRR (look: https://www.evidentlyai.com/ranking-metrics/mean-reciprocal-rank-mrr)
- NDCG (look: https://www.evidentlyai.com/ranking-metrics/ndcg-metric)

Loss functions for MF

Known loss functions used for RS

- MSE (regression)
- cross-entropy (classification)

Probabilitic Matrix Factorization

- User prior: $P(\mathbf{P}) = \prod_u \mathcal{N}(\mathbf{p}_u|0, \sigma_U^2 I)$
- Item prior: $P(\mathbf{Q}) = \prod_i \mathcal{N}(\mathbf{q}_i|0, \sigma_I^2 I)$
- Rating probability: $R_{i,j} \sim \mathcal{N}(.|\mathbf{p_u} \times \mathbf{q_i}, \sigma^2)$,
- 1_{ui} - indicates if user i has interacted with item j
- Likelihood:

$$p(R|\mathbf{P}, \mathbf{Q}, \sigma^2) = \prod_{u,i} \left[\mathcal{N}(r_{ui}|\mathbf{p_u} \times \mathbf{q_i}, \sigma^2)\right]^{1_{ui}}$$

- Posterior:

$$p(R|\sigma_U^2, \sigma_V^2, \sigma^2) = \prod_{u,i} \left[\mathcal{N}(r_{ui}|\mathbf{p_u} \times \mathbf{q_i}, \sigma^2)\mathcal{N}(\mathbf{p_u}|0, \sigma_U^2 I)\mathcal{N}(\mathbf{q_i}|0, \sigma_V^2 I)\right]^{1_{ui}}$$

Ruslan Salakhutdinov and Andriy Mnih. 2007. Probabilistic Matrix Factorization. In Proceedings of the 21st International Conference on Neural Information Processing Systems (NIPS'07). Curran Associates Inc., Red Hook, NY, USA, 1257–1264.

Loss function

Maximizing the log-posterior over movie and user features with hyperparameters (i.e. the observation noise variance and prior variances) kept fixed is equivalent to minimizing the sum-of-squared-errors objective function with L_2 regularization terms:

$$E = \frac{1}{2} \sum_{u,i} 1_{ui} (r_{ui} - \mathbf{p_u} \times \mathbf{q_i})^2 + \frac{\lambda_U}{2} \sum_u \|\mathbf{p_u}\|^2 + \frac{\lambda_I}{2} \sum_i \|\mathbf{q_i}\|^2$$

where $\lambda_U = \sigma^2/\sigma_U^2$, $\lambda_V = \sigma^2/\sigma_V^2$.

Optimization: gradient descent in $\mathbf{P}$ and $\mathbf{Q}$.

Scales linearly with number of observations.

Additional Tricks

- Scale Ratings to $[0, 1]$: $t(x) = (x - 1)/(K - 1)$
- Scale Dot-products to $[0, 1]$: $g(x) = 1/(1 + \exp(-x))$

- **Constrained version**:

$$\mathbf{p_u} = \tilde{\mathbf{p}}_u + \frac{\sum_i 1_{ui} W_i}{\sum_i 1_{ui}}$$

where $\tilde{\mathbf{p}}_u$ is the user-specific part and $W \in R^{D \times M}$ is the item specific part.

Bayesian Personalized Ranking

Idea:

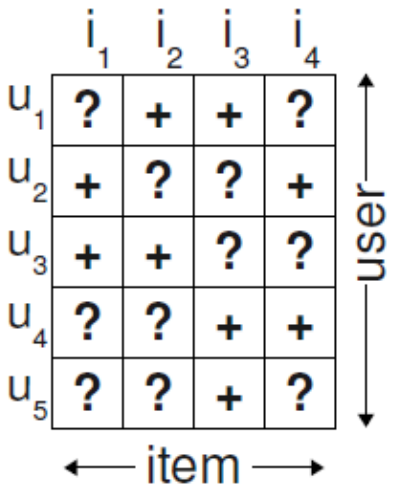
Provide a ranking of items – total order on all items:

$$\forall i, j \in I : i \neq j \Rightarrow i \geq_u j \vee j \geq_u i \quad \text{(totality)}$$
$$\forall i, j \in I : i \geq_u j \wedge j \geq_u i \Rightarrow i = j \quad \text{(antisymmetry)}$$
$$\forall i, j, k \in I : i \geq_u j \wedge j \geq_u k \Rightarrow i \geq_u k \quad \text{(transitivity)}$$

Intuition: optimize for correctly ranking item pairs instead of scoring single items

Rendle, S., Freudenthaler, C., Gantner, Z., & Schmidt-Thieme, L. (2009, June). BPR: Bayesian personalized ranking from implicit feedback.

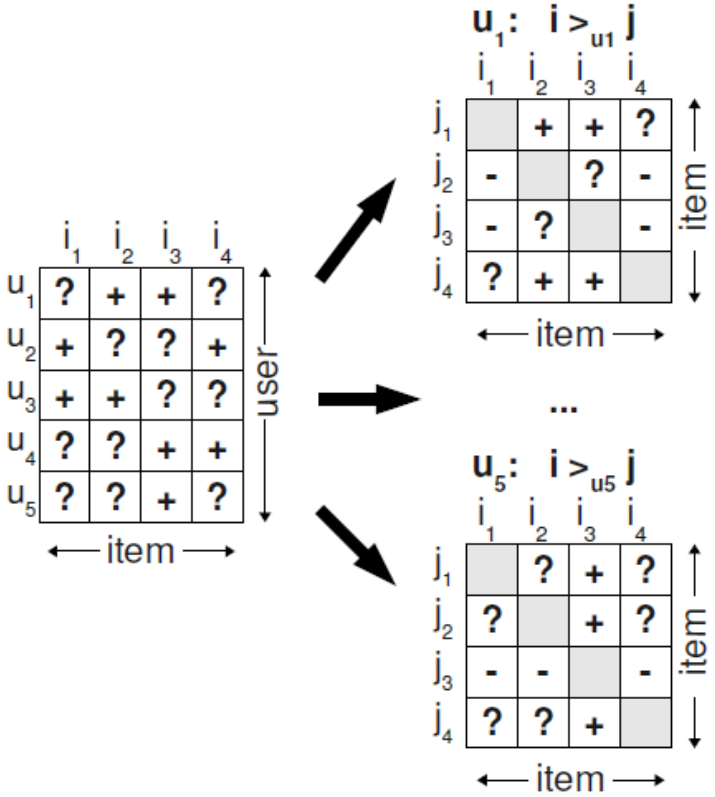
Data Representation:



$$D_S := \left\{(u, i, j) \mid i \in I_u^+ \wedge j \in I \setminus I_u^+\right\}$$

triples: $(user, positive, negative)$

Example



Loss function

$$\prod_{u \in U} p(\geq_u | \Theta) = \prod_{(ui,j) \in D_S} p(i \geq_u j | \Theta)$$

$$p(i \geq_u j | \Theta) := \sigma(\hat{x}_{ui,j}(\Theta))$$

$$\begin{aligned} BPR - OPT :&= \ln p(\Theta | \geq_u) \\ &= \dots \\ &= \sum_{(ui,j) \in D_S} \ln \sigma(\hat{x}_{ui,j}) - \lambda_\Theta \|\Theta\|^2 \end{aligned}$$

Uniformity & Alignment

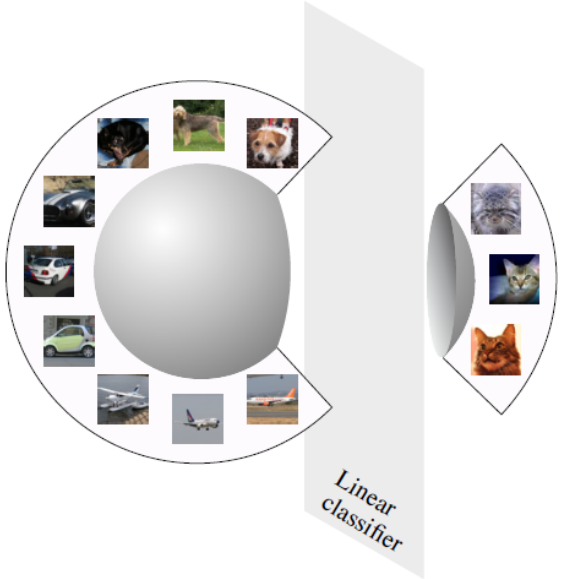


Figure 2: **Hypersphere:** When classes are well-clustered (forming spherical caps), they are linearly separable. The same does not hold for Euclidean spaces.

Tongzhou Wang and Phillip Isola. 2020. Understanding contrastive representation learning through alignment and uniformity on the hypersphere. In Proceedings of the 37th International Conference on Machine Learning (ICML'20), Vol. 119. JMLR.org, Article 921, 9929–9939.

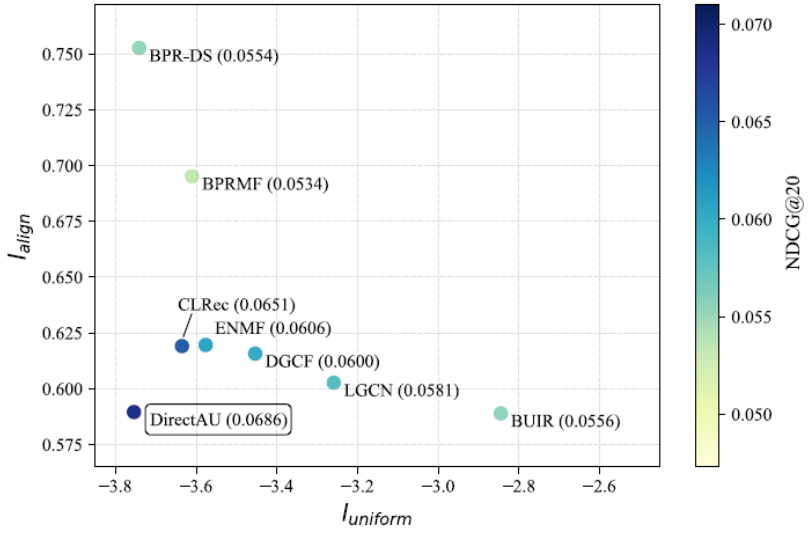
Uniformity & Alignment for RS

$$l_{align} = E_{(u,i)\sim positive} \|f(u) - f(i)\|^2$$

$$l_{uniformity} = \log E_{(u,u')} \frac{1}{2} e^{-2\|f(u)-f(u')\|^2} + \log E_{(i,i')} \frac{1}{2} e^{-2\|f(i)-f(i')\|^2}$$

Chenyang Wang, Yuanqing Yu, Weizhi Ma, Min Zhang, Chong Chen, Yiqun Liu, and Shaoping Ma. 2022. Towards Representation Alignment and Uniformity in Collaborative Filtering. In Proceedings of the 28th ACM SIGKDD Conference on Knowledge Discovery and Data Mining (KDD '22). Association for Computing Machinery, New York, NY, USA, 1816–1825. https://doi.org/10.1145/3534678.3539253

Relation to other metrics



Source: https://arxiv.org/abs/2206.12811

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